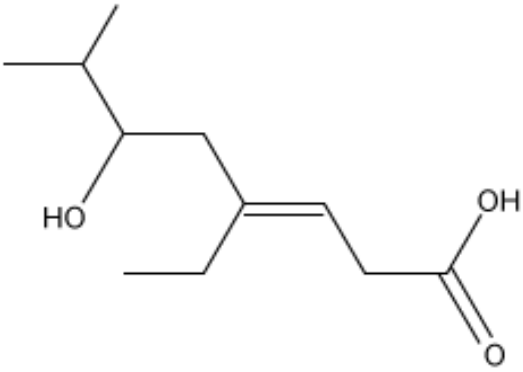
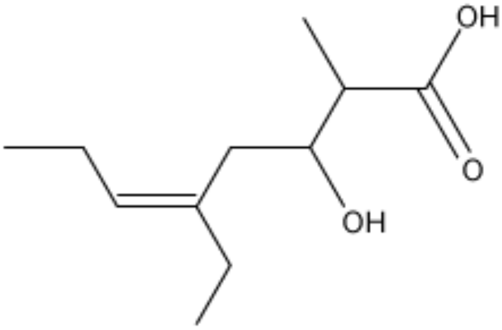
What is the structure of (*Z*)-4-ethyl-6-hydroxy-7-methyloct-3-enoic acid?

○ A.



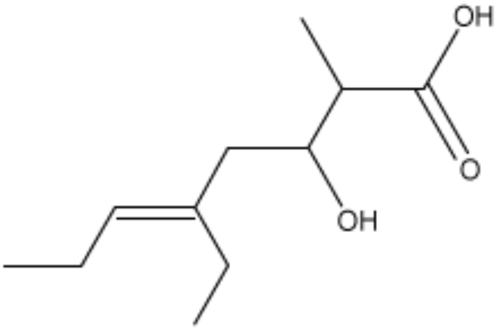
[12%]

○ B.



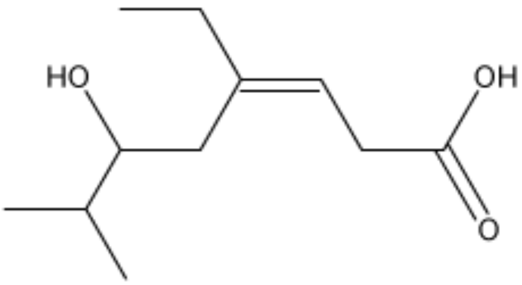
[10%]

○ C.



[5%]

✔ ◉ D.

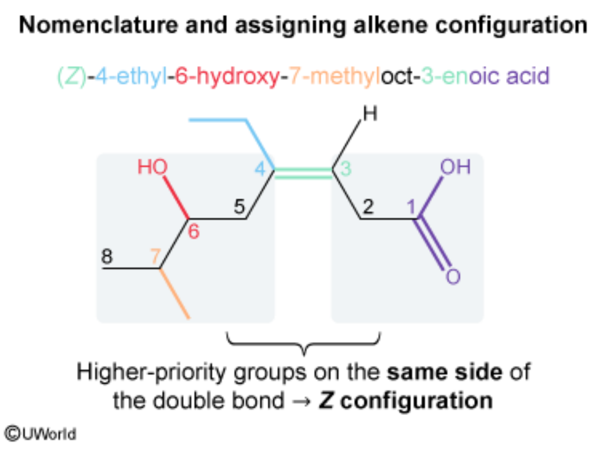


[71%]

Correct
70% answered correctly

Explanation:

微信Mai_ange



Alkenes are molecules containing double bonds that are di-, tri-, or tetra-substituted and are classified as *cis/trans* or *E/Z* based on priority rules. If the **higher-priority groups** are on **opposite sides** of the double bond, the alkene is ***E*** whereas the alkene is ***Z*** if the higher-priority groups are on the **same side** of the double bond.

For compounds with multiple functional groups, the highest-ranking group is used as the basis for naming the compound; it determines the suffix (ending) of the compound name, with all other functional groups named as substituents. The longest continuous carbon chain in the structure that includes the highest-priority functional group is numbered to give the highest-priority group the lowest possible number. Stereocenters (ie, chiral centers, alkenes) are listed parenthetically at the beginning of the compound name.

Based on the name given in the question, the suffix *–oic acid* indicates that a carboxylic acid is the highest-priority group. The prefix *oct–* denotes the longest continuous carbon chain containing the carboxylic acid consists of eight carbons that are numbered so the carboxylic acid carbon is position 1. Therefore, the substituents and their positions on the carbon chain are:

- **4-ethyl**: a two-carbon alkyl substituent at carbon 4
- **6-hydroxy**: a hydroxyl (–OH) group on carbon 6
- **7-methyl**: a one-carbon alkyl substituent at carbon 7
- **3-en**: an alkene at carbon 3 in the *Z* configuration

Choice D is the only structure that matches the given IUPAC name.

(Choice A) The higher-priority groups are on opposite sides of the double bond (ie, an *E* configuration instead of *Z*).

(Choice B) The indicated positions of the alkene and substituents are incorrect. The chain was incorrectly numbered giving the carboxylic acid carbon the highest instead of the lowest position.

(Choice C) The alkene has an *E* configuration, and the chain was incorrectly numbered giving the carboxylic acid carbon the highest position.

Educational objective:

An alkene is classified *E* if the higher-priority substituents are on opposite sides of the double bond whereas the alkene is classified *Z* when the higher-priority substituents are on the same side of the double bond. Names of compounds are based on functional group ranking and continuous carbon chain length.

Time spent: 0 sec	**QID**:403109
Subject: Organic Chemistry	**System:** Introduction to Organic Chemistry